**Client.c**

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#define PORT 8080
int main()
{
   int sockfd;
   char buffer[1024];
   struct sockaddr_in servaddr;
   sockfd = socket(AF_INET, SOCK_DGRAM, 0);
   servaddr.sin_family = AF_INET;
   servaddr.sin_port = htons(PORT);
   inet_pton(AF_INET, "127.0.0.1", &servaddr.sin_addr);
   char *msg = "TIME REQUEST";
   printf("Sending time request to server...\n");
   sendto(sockfd, msg, strlen(msg), 0,
         (const struct sockaddr *)&servaddr, sizeof(servaddr));
   int len = sizeof(servaddr);
   int n = recvfrom(sockfd, buffer, 1024, 0,
                  (struct sockaddr *)&servaddr, &len);
   buffer[n] = '\0';
   printf("Server System Time: %s\n", buffer);
  close(sockfd);
   return 0;
}
```

**Server.c**

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <time.h>
#define PORT 8080
int main()
{
   int sockfd;
   char buffer[1024];
   struct sockaddr_in servaddr, cliaddr;
   sockfd = socket(AF_INET, SOCK_DGRAM, 0);
   servaddr.sin_family = AF_INET;
   servaddr.sin_addr.s_addr = INADDR_ANY;
   servaddr.sin_port = htons(PORT);
   bind(sockfd, (const struct sockaddr *)&servaddr, sizeof(servaddr));
   printf("Current Time Server (UDP) ready on port %d...\n", PORT);
   while (1)
   {
      int len = sizeof(cliaddr);
      recvfrom(sockfd, buffer, 1024, 0,
            (struct sockaddr *)&cliaddr, &len);
      if (fork() == 0)
      {
         time_t t = time(NULL);
         char *time_str = ctime(&t);
         sendto(sockfd, time_str, strlen(time_str), 0,
```

```
            (const struct sockaddr *)&cliaddr, len);
        exit(0);
      }
   }
   return 0;
}
```

**Output**

**Client**

Sending time request to server...

Server System Time: Fri Sep 18 09:36:42 2026

**Server**

Current Time Server (UDP) ready on port 8080...